existence of dark matter and dark energy. As we see it, theoretical physics risks becoming a no-man's-land between mathematics, physics and philosophy that does not truly meet the requirements of any." Straying from Popper's dictum, they warned, was a "drastic step" with potentially dire consequences. "This battle for the heart and soul of physics is opening up at a time when scientific results—in topics from climate change to the theory of evolution—are being questioned by some politicians and religious fundamentalists. Potential damage to public confidence in science and to the nature of fundamental physics needs to be contained by deeper dialogue between scientists and philosophers."

Yet if Ellis and Silk had bothered to engage in such a dialogue before writing their editorial, they would have learned that Popper's work hasn't been taken seriously by philosophers of science for decades, and with good reason. The idea that falsifiability marks the boundary of science is vulnerable to the same arguments that made the verification theory of meaning untenable, which we saw in Chapter 8. Just as individual beliefs can't be verified, as Quine pointed out in "Two Dogmas of Empiricism," individual theories can't be falsified either, for much the same reason. Say Karl Popper's remote control isn't turning his television on, and he theorizes that the batteries in the remote are dead. So Popper runs down to the corner store, buys new batteries, and pops them into his remote. But the remote still doesn't turn the TV on. "Aha!" Popper exclaims. "My theory has been falsified!" But it ain't necessarily so. Despite the fact that the remote still isn't working, it's still possible that the old batteries were dead. Maybe the new batteries are dead too. Maybe a rat gnawed through the TV's power cable while Popper was at the corner store. Maybe the laws of physics actually change depending on where you are, and while Popper was at the store, the Solar System moved through its orbit around the center of the Milky Way and entered a patch of space where the laws of electromagnetism that govern the behavior of batteries in remote controls are different. The problem is that Popper's "dead battery theory" of his remote control doesn't actually make any predictions on its own: it only makes predictions in conjunction with a huge number of other basic assumptions that Popper has made about the functioning of the world. So Popper is wrong: his theory was not falsified. When the remote continued not to work, he could have rejected his theory about the dead batteries, but